*November 5, 2026*

# U-space USSP-USSP Exchanges for Network Identification SWIM Service Description Document

| General Service Description | |
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| **Service description identification**<br><br>**SWIM-SERV-030** | **Title:** U-space USSP-USSP Exchanges for Network Identification SWIM Service Description<br><br>**Edition:** V1.1.1<br><br>**Reference Date:** 2026-11-05 |
| **Service identification**<br><br>**SWIM-SERV-040** | **Name:** U-space USSP-USSP Exchanges for Network Identification Service<br><br>**Version:** V1.1 |
| **Service abstract**<br><br>**SWIM-SERV-050** | The network identification service is one of the four mandatory services under the Commission Implementing Regulation (EU) 2021/664. The network identification service provides the registration number of a UAS operator, the serial number of an unmanned aircraft, and live flight data of the UAS. It enables the sharing of information with any of the authorised users listed in Article 8(4)(b) of Regulation (EU) 2021/664. Authorised users will be made aware of the geographical position, route course and emergency status, flight level, and type of the UAS, among other data elements. Based on the information provided by the UAS operators, USSPs share and consolidate UAS flight data among themselves and can, therefore, support traffic information when needed.<br><br>This service implementation is based on the ASTM F3411-22a standard, identified as an applicable standard in U-space accepted means of compliance. |
| **Service Provider and Contact Information** | |
| **Service Provider**<br><br>**SWIM-SERV-060** | **Name:** U-space Service Provider (see Operations Committee list for participating providers)<br><br>**Description of organization:** U-space Service Provider refers to an organisation that is certified by a competent authority to provide U-space services in the U-space airspace (Ref: GM1a Article 7 U-space service providers) |
| **Service Characteristics** | |

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| **Geographical extent of information**<br><br>**SWIM-SERV-090** | Geographical coverage of exchanged information service payload is available through the Discovery and Synchronization Service under ASTM F3411-22a Section A2. |
| **Service categories**<br><br>**SWIM-SERV-100** | **Service category:** Flight information |
| **Service lifecycle information**<br><br>**SWIM-SERV-110** | **Lifecycle stage and if applicable qualifying dates:** As published by the Operations Committee. |
| **Service standard reference**<br><br>**SWIM-SERV-120** | **Service standard reference:** Service Standard Reference: Standard Specification for Remote ID and Tracking (ASTM F3411-22a)<br><br>**Implemented options of the service standard:**<br>Net-RID Service Providers and Net-RID Display Providers must provide the following optional fields in ASTM F3411-22a Table 1:<br>● UAS ID - 1. Serial Number<br>● Operational Status<br>● Operator ID<br>● Height<br>● Height Type<br>● Operator Latitude<br>● Operator Longitude<br><br>**Net-RID Display Providers must provide all required fields in ASTM F3411-22a Table 1.**<br><br>**Addition to the service standard**<br>Per U-space regulation (EU) 2021/664, AMC1 to Article 8(2)(c), USSPs must convert the altitudes above the WGS 84 ellipsoid exchanged with the ASTM F-3411-22A standard to altitude above mean sea level (MSL) using either the EGM96 or the EGM2008 geoid model as the definition of mean sea level before providing it.<br><br>Discovery and synchronization service configuration process - as defined by InterUSS DSS Pooling instructions on joining an existing pool with new instance.<br><br>**DSS deployment requirements (minimum)**<br>Each USSP joining the DSS pool shall deploy its DSS instance with the following configuration:<br><br>● DSS version: v0.23.0<br>● Datastore: CockroachDB |

<table>
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- Datastore version: v24.1.3
- Number of nodes: 3
- Configurable flags: None

Each DSS node shall meet the following minimum specification:

- Dedicated vCPUs: 4
- Memory (RAM): 16 GB
- Storage: 20 GB minimum
- Network latency between DSS nodes in the pool should be kept below 20 ms round-trip 95% of the time

**DSS clusters**

<table>
<tr><th>Environment</th><th>Cluster name</th></tr>
<tr><td>Test (Gate 2)</td><td>uspace-zurich-test</td></tr>
<tr><td>Production (Gate 3)</td><td>uspace-zurich-ops</td></tr>
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Note: These clusters may be shared with other U-space services that rely on the DSS. A separate cluster is not required for the Network Identification Service.

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<tr><th colspan="2">High-level Description of Service Offer</th></tr>
<tr><td>Operational environment<br><br>SWIM-SERV-130</td><td>

Refer to the following documents for operational needs addressed by the service and the capabilities offered by the service:

- Commission Implementing Regulation (EU) 2021/664 - Preamble
- ASTM F3411-22a Sections 4.8 and A2 capabilities:
  - USS-DSS Interfaces
  - USS-USS Interfaces
  - DSS-DSS Synchronization

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<tr><td>Service functions<br><br>SWIM-SERV-140</td><td>

Refer to the following documents for functions offered by the service and their real world effects.

- Commission Implementing Regulation (EU) 2021/664 Article 8
- Article 8 Accepted Means of Compliance and Guidance Material to Regulation (EU) 2021/664 — Issue 1
- ASTM F3411-22a Section A2 functions are provided for the following capabilities:
  - USS-DSS Interfaces
  - USS-USS Interfaces
  - DSS-DSS Synchronization

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<tr><th colspan="2">Limitations and Constraints on Using the Service</th></tr>
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| **Service access and use conditions**<br><br>**SWIM-SERV-150** | Refer to the EU USSP U-space Data Sharing and Governance Agreement for information on legal constraints, service policies, and service consumption constraints |
| **Security constraints**<br><br>**SWIM-SERV-160** | Refer to the following documents for security requirements.<br>● For field testing (U-space v1), security compliance shall be demonstrated either by:<br>○ holding a valid USSP certification in accordance with Regulation (EU) 2021/664, or<br>○ holding an ISO/IEC 27001 certification and being in the process of obtaining USSP certification, or<br>○ complying to comparable security industry standards (such as IEC 62443) and being in the process of obtaining USSP certification.<br><br>This recognition is a transitional measure limited to field testing and is not applicable to full U-space deployment.<br><br>● ASTM F3411-22a Sections 5.5.4 and A2 (relevant requirements are listed below):<br>○ NET0210<br>○ NET0220<br>○ DSS0010<br>○ DSS0020<br>○ DSS0110<br>○ DSS0120<br><br>*Per ASTM F3548-21, Appendix X1 - Reference architecture for interoperability security controls, The X1.1 Base Deployment: Access Tokens with Audience Claims is implemented. Authentication and Authorization for communication between USS-USS and USS-DSS shall be facilitated by a common interoperability token exchanger service supporting OAuth 2.0.* |
| **Quality Aspects** | |
| **Quality of service**<br><br>**SWIM-SERV-180** | Refer to the following documents for quality requirements.<br>● EU USSP U-space Data Sharing and Governance Agreement - Annex XX Service Level Agreement for information on quality of service |
| **Service validation information**<br><br>**SWIM-SERV-200** | Service will be verified using the InterUSS Automated Test Baseline prior to software deployment using a test configuration with a test baseline identifier of TB-2d02f9c. |
| **Behavior of the Service** | |
| **Application** | Refer to the following sections in ASTM F3411-22a for information |

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| **message exchange pattern**<br><br>**SWIM-SERV-210** | about the application message exchange pattern(s) used by the service.<br>* A2 - Network Remote ID interoperability requirements, APIs, and testing<br>* A4 - USS-DSS and USS-USS OpenAPI YAML Description |
| **Service behavior**<br><br>**SWIM-SERV-220** | Refer to the attached Compliance Matrix, *U-space Article 8 Network Identification Service Compliance Matrix V1.0* for service performance and verification requirements |
| **Service Implementation and Structural Details** | |
| **Service interfaces**<br><br>**SWIM-SERV-240** | Refer to the following sections in ASTM F3411-22a for information about service interfaces.<br>* A2 - Network Remote ID interoperability requirements, APIs, and testing<br>* A4 - USS-DSS and USS-USS OpenAPI YAML Description |
| **Service interface protocols and data format**<br><br>**SWIM-SERV-260** | Refer to the following sections in ASTM F3411-22a for information about service interface protocols and data format.<br>* A2 - Network Remote ID interoperability requirements, APIs, and testing<br>* A4 - USS-DSS and USS-USS OpenAPI YAML Description |
| **Service operations**<br><br>**SWIM-SERV-270** | Refer to the following sections in ASTM F3411-22a for information about service operations.<br>* A2 - Network Remote ID interoperability requirements, APIs, and testing<br>* A4 - USS-DSS and USS-USS OpenAPI YAML Description |
| **Service messages**<br><br>**SWIM-SERV-280** | Refer to the following sections in ASTM F3411-22a for information about messages exchanged by the service.<br>* A2 - Network Remote ID interoperability requirements, APIs, and testing<br>* A4 - USS-DSS and USS-USS OpenAPI YAML Description |
| **Information Aspects of the Service** | |
| **Information definition (minimum)**<br><br>**SWIM-SERV-290** | Refer to the following sections in ASTM F3411-22a for information about the exchanged information service payload.<br>* A2 - Network Remote ID interoperability requirements, APIs, and testing<br>* A4 - USS-DSS and USS-USS OpenAPI YAML Description |
| **Resources** | |
| **Machine-readable service interface definition**<br><br>**SWIM-SERV-320** | Refer to the following sections in ASTM F3411-22a for information about machine-readable service interface definition.<br>* A4 - USS-DSS and USS-USS OpenAPI YAML Description |

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| **Abbreviations and acronyms**<br><br>**SWIM-SERV-350** | Refer to the following sections in ASTM F3411-22a for abbreviations and acronyms:<br>* 3 - Terminology<br>* A2 - Network Remote ID interoperability requirements, APIs, and testing<br>* A4 - USS-DSS and USS-USS OpenAPI YAML Description |